Domain-Specific Lossless Compression of 16-bit Astronomical Images

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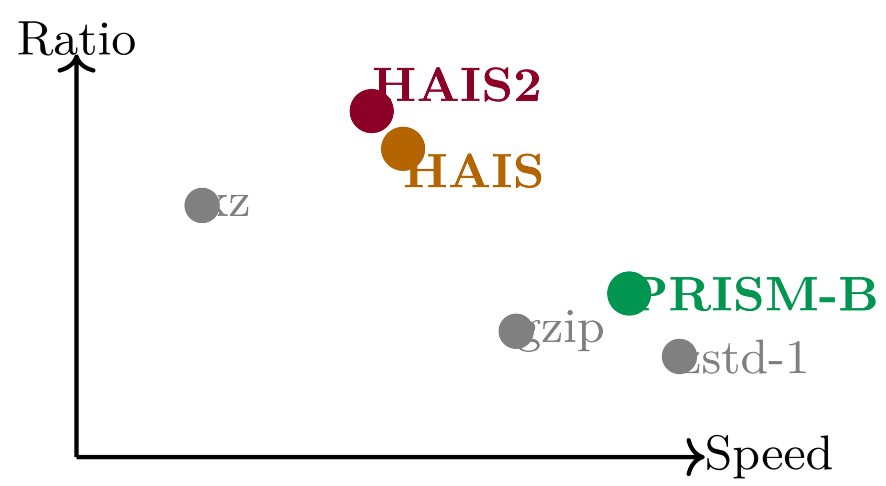
1. Motivation & Problem

Modern telescopes produce **16-bit raw image streams** where storage and transmission bandwidth are critical. General-purpose compressors (gzip, xz, zstd) treat pixels as bytes, ignoring spatial structure.

Our goal: design lossless codecs that exploit the unique statistics of astronomical imagery:

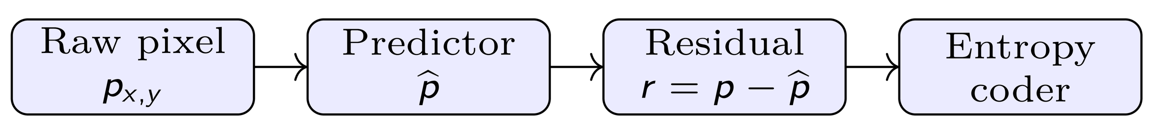
- Smooth, near-uniform backgrounds (high spatial correlation)
- Sparse bright point sources (sparse residuals)
- 16-bit dynamic range — byte-split creates correlated pairs
- Fixed 1500×1500-pixel frames (repeatable block geometry)

Design axes: compression ratio ↔ speed ↔ simplicity. We present **three codecs** on this Pareto frontier.



2. Common Architecture

All three codecs share a **predict-residualise-encode** pipeline:



Residuals are **zigzag-mapped** to non-negative integers and split into a high byte (hi) and low byte (lo) before entropy coding. The codecs differ in *predictor* and *entropy coder*.

3. PRISM-Block — Fast Codec

Predictor — GAP (Gradient-Adaptive):

$$\hat{p} = \frac{(\delta_v + 1)W + (\delta_h + 1)N}{\delta_v + \delta_h + 2}, \qquad \begin{aligned} \delta_h &= |W - NW| \\ \delta_v &= |N - NW| \end{aligned}$$

Uses only causal neighbours (W, N, NW), with edge cases biased toward the stronger horizontal or vertical gradient.

Entropy coder — Adaptive Range Coder:

- Fenwick-tree frequency model (256 symbols, rescales at 2^{16})
- **9 contexts:** 1 for hi-byte; 8 for lo-byte conditioned on local residual magnitude $m_W + m_N$
- LZMA-style carry-propagation encoder

Parallelism — Independent Row Blocks:

- Image split into 150-row horizontal bands
- Each band encoded independently ⇒ zero inter-thread sync
- `std::thread` pool, as many threads as CPU cores

Micro-optimisations:

- First-row and first-pixel cases peeled from inner loop
- Context lookup via precomputed `LO_CTX_TAB[1024]`
- Works on *any* image dimensions via `infer_dims()`

	Compress	Decompress
Avg. time	**35 ms**	**46 ms**
Avg. ratio	3.51 b/B	—

4. HAIS2 — All-Rounder Codec

Block-Adaptive Multi-Predictor:
Each $\mathrm{BS} \times \mathrm{BS}$ block independently selects the predictor with minimum estimated entropy:

Mode	Name	Predictor	Overhead
0	raw	identity	0 B
1	avg	$(W + N + NW)/3$	0 B
3	mean	global mean μ	0 B
4	LS4	$\mathrm{LS}(W, N, NW, 1)$	16 B
6	LS5	$\mathrm{LS}(W, N, NW, NE, NN)$	20 B
9	LS6	$\mathrm{LS}(W, N, NW, WW, NN, 1)$	24 B

LS weights are solved per block. The estimator scores $H(\mathrm{hi}) + \frac{n_0}{N} H(\mathrm{lo}|h = 0) + \frac{n_1}{N} H(\mathrm{lo}|h > 0)$.

Entropy Coder — ANS/FSE:

- Static per-block frequency table (sparse: $5 \times \mathrm{nnz}$ bytes, dense: 1025 bytes)
- Hi and lo streams encoded separately
- Optional *ctx* mode: lo split into $h = 0$ / $h > 0$ sub-streams

Block size selection: largest $\mathrm{BS} \leq 512$ dividing both W and H with ≥ 4 blocks. For 1500×1500: $\mathrm{BS} = 500$ (9 blocks). Smaller blocks lose to FSE table overhead.

Predictor distribution on test set:

File type	Winner	Notes
Star field (A)	avg (9/9)	4.39 b/B
Smooth bg (C,D,G,H)	mean/LS6	2.41–2.73 b/B
High contrast (F)	LS6 (9/9)	6.11 b/B
Sparse stars (E)	raw+mean	2.57 b/B

6. Benchmark Results

8 test images, 1500×1500 pixels, 16-bit raw. Averages across all files.

Codec	Avg b/B ↓	Compress ↓	Decompress ↓	Notes
gzip -6	4.376	473 ms	61 ms	General-purpose
bzip2 -9	3.579	394 ms	234 ms	Block-BWT
xz -6	3.685	2458 ms	160 ms	Best general ratio
zstd -1	4.488	27 ms	15 ms	Speed reference
PRISM-Block	**3.510**	**35 ms**	**46 ms**	Fast adaptive RC
HAIS	**3.400**	**89 ms**	**98 ms**	Block FSE
HAIS2	**3.396**	**92 ms**	**102 ms**	LS6 + ctx estimator

All three of our codecs outperform gzip and xz in *both* ratio and speed. HAIS2 beats bzip2 in ratio while being 4× faster to compress and 2.3× faster to decompress.

5. HAIS2+ — Best Ratio (In Development)

Known residual gaps (from information theory analysis):

- File A: lag-1 autocorrelation of residuals after avg = **0.79** (raw = 0.95), so spatial structure remains.
- Hi/lo byte split overhead ≈ 22 KB total across 8 files
- File A: 8.77 b/B achieved vs 8.70 b/B theoretical

Planned improvements:

1. **CALIC-style bias correction** — per-context mean correction before entropy coding.
2. **16-bit FSE table** for hi==0 blocks — eliminates the byte-split overhead for smooth regions.
3. **Adaptive LS predictor** — online weight update using running covariance.

Why not GAP or MED? Analysis confirmed these never win on this data: astronomical images are either smooth (mean/LS wins) or star-field (avg wins). Including them adds latency with zero gain.

Current status: bias correction prototype tested; results pending. Expected gain: $\Delta \approx 0.02$–0.05 b/B on star-field images.

7. Conclusions

Key takeaways:

- Domain knowledge (spatial prediction, byte-split entropy) beats general-purpose compressors by 0.2–1.0 b/B while running faster.
- Predictor selection per block is the single highest-impact design choice; LS6 dominates on smooth and gradient regions.
- Residual autocorrelation (0.79 for star fields) remains the main open challenge, targeted by HAIS2+.

Recommendations by use case:

Scenario	**Codec**
Real-time acquisition	PRISM-Block
Archival storage	HAIS2
Maximum ratio	HAIS2+ (dev)

All code is original (C++17, no third-party compression libraries). Benchmark comparisons invoke external tools as separate processes.